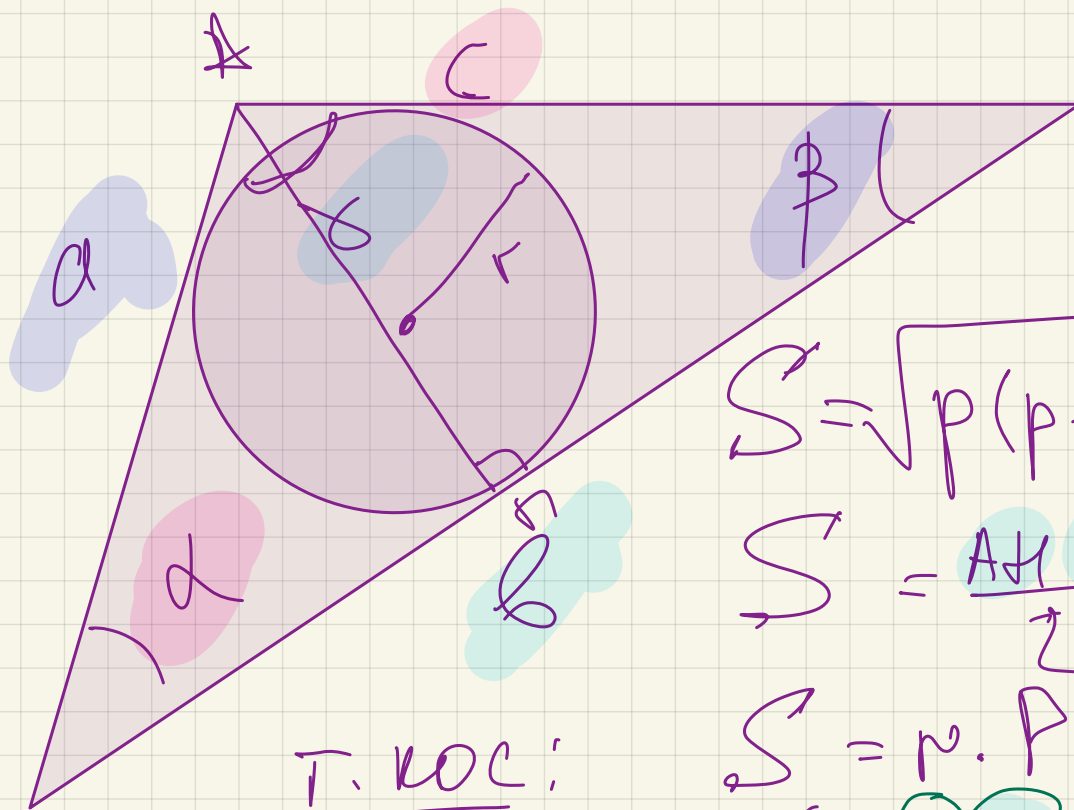




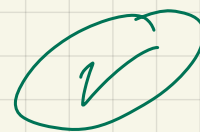


$$S = \sqrt{p(p-a)(p-b)(p-c)}$$

$$S = \frac{A \cdot h}{2} \cdot b$$

$$S = r \cdot P$$

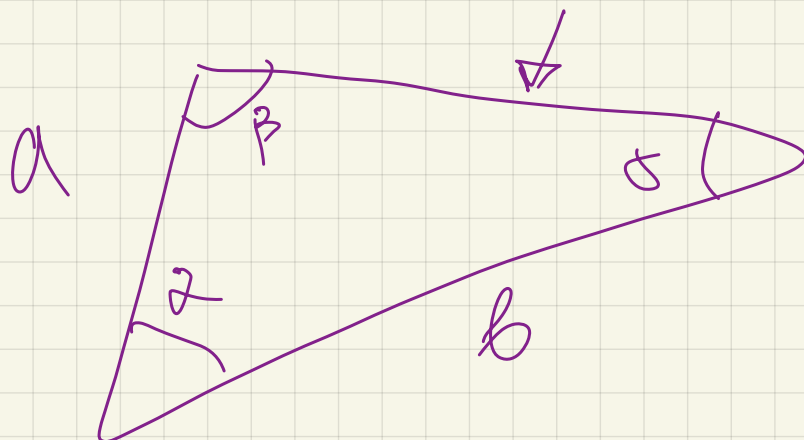
$$S = \frac{1}{2} ab \cdot \sin \alpha$$



$$a^2 + b^2 - 2ab \cos \alpha = c^2$$

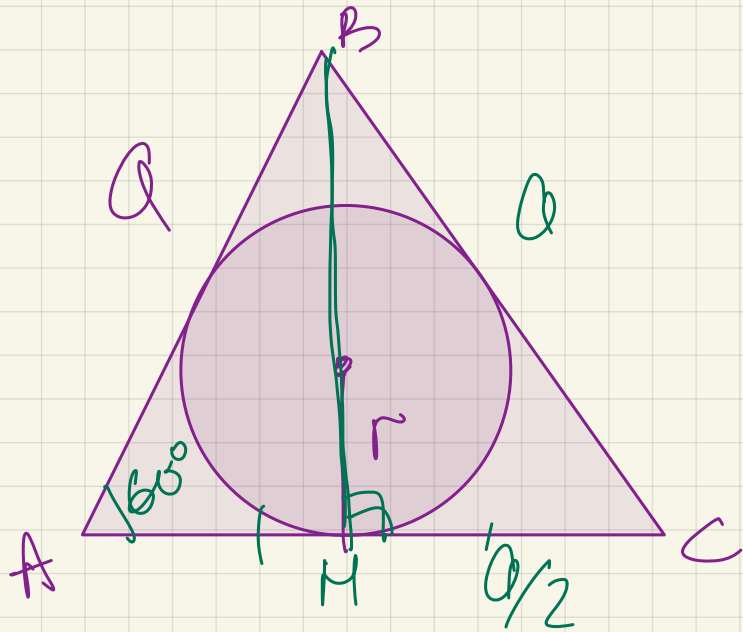
T. Sm

$$\frac{a}{\sin B} = \frac{b}{\sin \alpha} = \frac{c}{\sin \alpha} = 2R$$



$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta}$$

$$\sin \beta = \frac{b \sin \alpha}{a}$$



$$r ?$$

$$h = 45$$

$$r = ?$$

$$\sin 60 = \frac{\sqrt{3}}{2}$$

$$S = \frac{\sqrt{3}}{2} \cdot \frac{1}{2} a \cdot a = \frac{a^2 \sqrt{3}}{4}$$

$$p = \frac{3a}{2}$$

$$r = \frac{S}{p} = \frac{\frac{\sqrt{3}}{2} \cdot a}{\frac{3a}{2}} = \frac{a\sqrt{3}}{3}$$

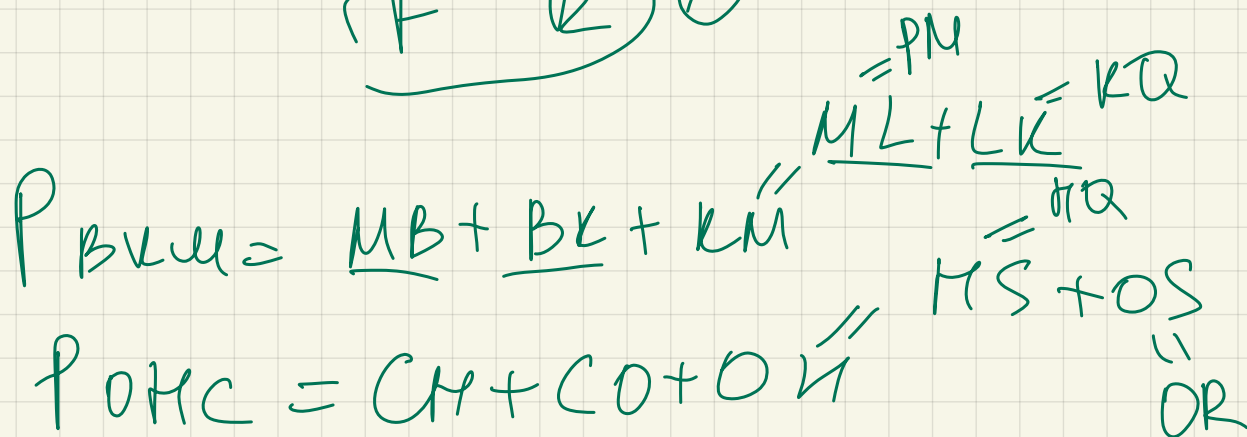
$$r = \sqrt{a^2 - \left(\frac{a}{2}\right)^2} = \frac{a^2 \sqrt{3}}{4} : \frac{3a}{2} = \frac{a^2 \sqrt{3} \cdot 2}{2^2 \cdot 3a}$$

$$h = \sqrt{a^2 - \frac{a^2}{4}} = \sqrt{\frac{4a^2 - a^2}{4}} = \sqrt{\frac{3a^2}{4}} = \frac{a\sqrt{3}}{2}$$

$$h = \frac{a\sqrt{3}}{2}$$

$$r = \frac{a\sqrt{3}}{6}$$

$$\boxed{\frac{h}{r} = 3}$$



$$P_{BKM} = \underline{MB} + \underline{BK} + K''$$

$$P_{OHC} = C_H + C_O + O''$$

$$P_{ANF} = AN + \frac{NF}{1} + AF$$

$$P_{ABC} = A^N + N^P + P^M + M^P +$$

$$+ K^K + K^Q + Q^M + M^C +$$

$$C^O + O^R + R^F + F^A$$

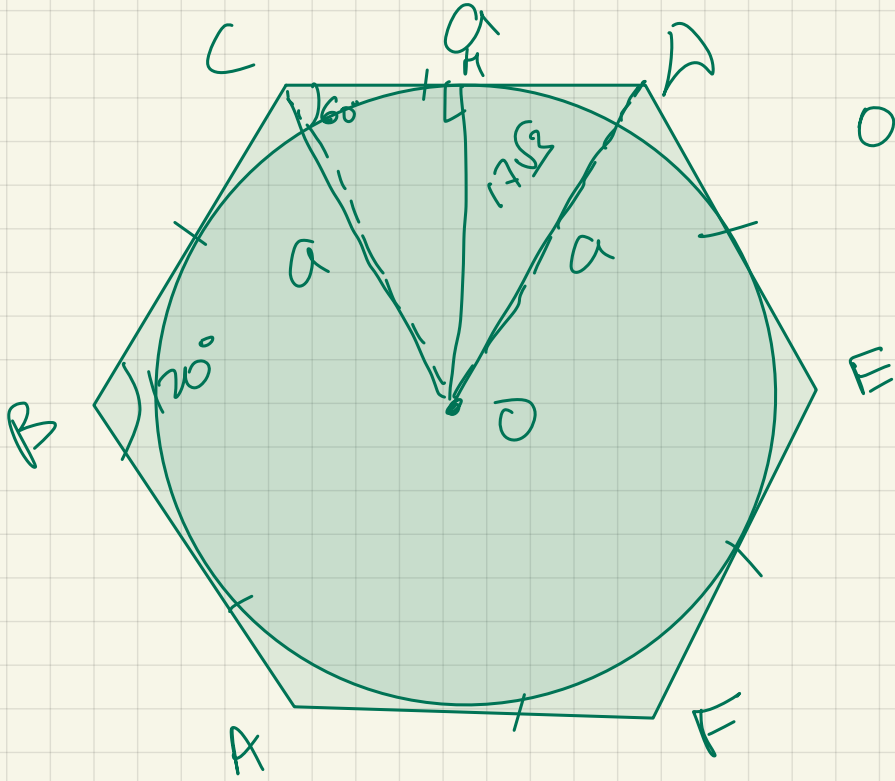
$$P_{ANF} = AN + AF + NP + RF$$

$$\text{Pore C} = \text{CH} + \text{CO} + \text{H}_2 + \text{O}_2$$

$$P_{ORM} = \mu_B + B_X + \mu_M + k_Q$$

$$P_{ABC} = P_{\emptyset MK} + P_{KCD} + P_{ANF} =$$

$$8 + 23 + 78 = 109$$



$$OH = 17\sqrt{3}$$

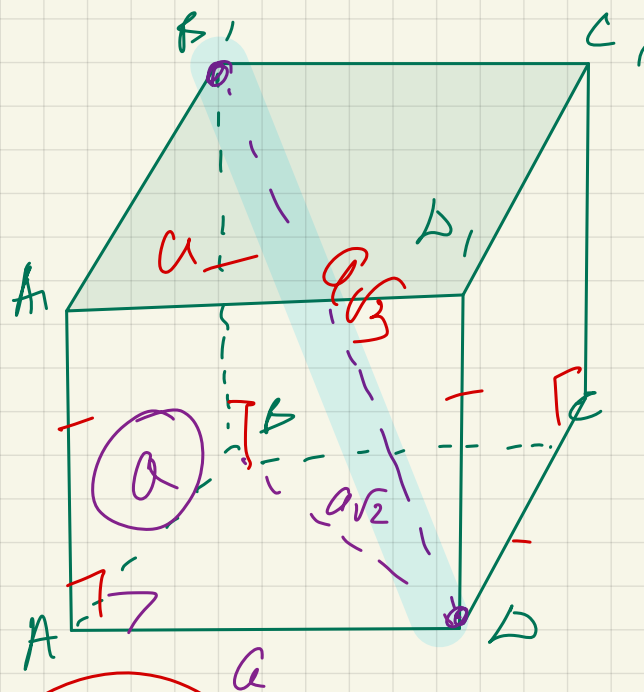
Q?

$$\cos 60 = \frac{\sqrt{3}}{2}$$

$$\frac{bH}{CO} = \ln C = \frac{\sqrt{3}}{2}$$

$$\frac{DH}{\sqrt{3}} \cdot 2 = 60$$

$$\frac{17\sqrt{3} \cdot 1}{\sqrt{3}} = \underline{\underline{34}}$$



$$BA = \sqrt{a^2 + a^2} = \underline{a\sqrt{2}}$$

$$d = \underline{BA} = \sqrt{BA^2 + BA'^2} \quad BA^2 + BA'^2 = BA'^2$$

$$\sqrt{2a^2 + a^2} = \sqrt{3a^2} = \underline{a\sqrt{3}}$$

$$S = 6a^2$$

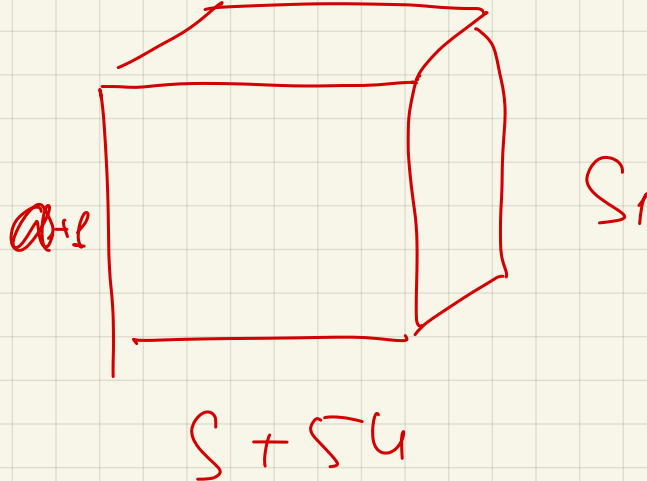
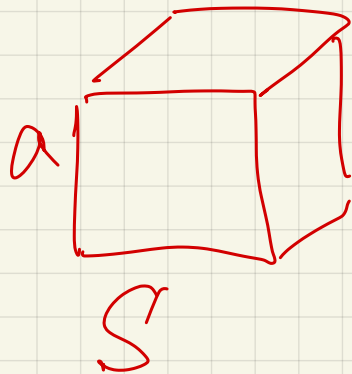
$$V = a^3$$

$$S' = 18 = 6a^2 \rightarrow a = \sqrt{3}$$

$$d? \quad d = \sqrt{3} \cdot \sqrt{3} = 3$$

$$V = 8$$

$$S?$$



a?

$$S = 6a^2$$

$$S_1 = 6(a+1)^2 = 54$$

$$(a+1)^2 = \frac{54}{6} = 9$$

$$(a+1)^2 = 9$$

$$a^2 + 1 + 2a$$

$$(a+1)(a+1) \Rightarrow a^2 + a + a + 1 = a^2 + 2a + 1$$

$$a^2 + 2a + 1 = 9$$

$$a^2 + 2a - 8 = 0$$

$$a \rightarrow (a+1)$$

$$V \rightarrow (V+19)$$

$$V = a^3$$

$$(a+1)^3 = V+19$$

$$(a+1)^3 = a^3 + 19$$

$$(a^2 + 2a + 1)(a+1) = a^3 + a^2 + 2a^2 + 2a + a + 1 =$$
$$\underline{\underline{a^3 + 3a^2 + 3a + 1}}$$

$$\cancel{a^3} + 3a^2 + 3a + 1 = \cancel{a^3} + 19$$

$$3a^2 + 3a - 18 = 0$$

$$a = ?$$